

# A Comparative study on Total Ionizing Dose Effects of 100-V Split-gate Trench VDMOS and Conventional Trench VDMOS

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**Abstract**—In this paper, we present a comparative study on the effects of Total Ionizing Dose (TID) radiation on the Split-Gate Trench VDMOS (SGT-MOS) and the conventional Trench VDMOS (T-MOS), both with equal rated breakdown voltages of 100 V, using experimental methods. The experimental findings reveal that the same degradation trends in the threshold voltage ( $V_{th}$ ) were found for both types of devices. However, the breakdown voltage ( $BV$ ) of SGT-MOS exhibits greater sensitivity to TID radiation compared to the T-MOS. Utilizing Sentaurus TCAD, we analyze and discuss the variances in  $BV$  shifting behavior between the SGT-MOS and T-MOS. The differences in  $BV$  shifting behavior between SGT-MOS and T-MOS can be attributed to the positive oxide trapped charges induced by TID, which could significantly affect the electric field distribution for SGT-MOS, but not for T-MOS. To address this issue, three methods are proposed to enhance the TID tolerance for the SGT-MOS and T-MOS. These methods include avoiding high-temperature processes, reducing the thickness of the oxide layer, and decreasing the doping in the drift layer.

**Keywords**—Total ionizing dose (TID), Split-gate trench VDMOS (SGT-MOS), Conventional Trench VDMOS (T-MOS), threshold voltage ( $V_{th}$ ), breakdown voltage ( $BV$ )

## I. INTRODUCTION

Due to its high cell density, low power loss and high current density, the conventional trench VDMOS (T-MOS) has

demonstrated great potential in aeronautics and astronautics applications, especially in the low-voltage range [1],[2]. The split-gate trench VDMOS (SGT-MOS), a special kind of T-MOS, achieves superior electrical performance benefiting from a deep trench structure with a split gate [3]-[5].

However, the utilization of SGT-MOS and T-MOS in aerospace environments is constrained by radiation effects. The total ionizing dose (TID) effect is one of the most common types of radiation damage in power devices used in space equipment [6],[7]. Under the influence of ionizing radiation, SGT-MOS and T-MOS are susceptible to TID effects, resulting in device performance degradation and even failure. This is primarily due to the positive oxide trapped charges and the interface charges induced by TID, which lead to a shift in threshold voltage ( $V_{th}$ ), an increase in leakage current, and a decrease in breakdown voltage ( $BV$ ) [8]-[11]. Although recent studies have explored planar gate devices, there are few investigations into the degradation behavior and TID radiation-hardening technologies of SGT-MOS and T-MOS [12]-[15].

In this paper, we report, for the first time, a comparative experimental study on TID effects of 100 V-class SGT-MOS and T-MOS. Section II introduces the sample devices of SGT-MOS and T-MOS, as well as the experimental conditions. The test results are then analyzed and discussed in comparison with data and simulations in Section III. Finally, we propose three

approaches to improve the TID tolerance on  $V_{th}$  and  $BV$  for both SGT-MOS and T-MOS.

## II. DEVICES AND EXPERIMENTS

Fig.1 (a) and (b) illustrate the cross-sectional views of the n-channel conventional trench VDMOS (T-MOS) and split-gate trench VDMOS (SGT-MOS), respectively. Both devices are manufactured with 0.35- $\mu\text{m}$  process and designed to achieve the same rated breakdown voltage of 100 V. The threshold voltage of SGT-MOS and T-MOS are 2.8 V and 2.3 V, respectively. The scanning electron microscope photograph for both devices are shown in Fig.1 (c) and (d). Compared to T-MOS, the SGT-MOS has a more complex trench structure, consisting of the gate oxide layer, isolation oxide layer, and split-gate oxide layer. Owing to its unique trench structure, the drift layer doping concentration of SGT-MOS can reach  $3.5 \times 10^{16} \text{ cm}^{-3}$ , which is significantly higher than  $2.5 \times 10^{15} \text{ cm}^{-3}$  of T-MOS. The main structure parameters for SGT-MOS and T-MOS are presented in Table I.

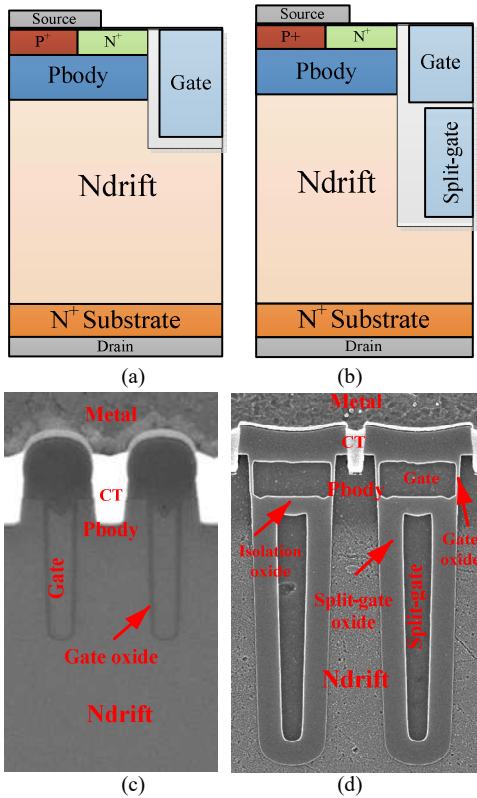


Fig. 1. The cross-sectional views and SEM photographs of T-MOS (a), (c) and SGT-MOS (b), (d).

For total ionizing dose tests, SGT-MOS and T-MOS are assembled and mounted into regular SMD-0.5 type packages. All samples are electrically tested before irradiation. A  $\text{Co}^{60}$  source at the China Academy of Engineering Physics, with a dose rate of 50 rad(Si)/s, is used for these TID tests. The devices are irradiated under different bias conditions, including on-state bias ( $V_{GS} = 12 \text{ V}$ ,  $V_{DS} = 0 \text{ V}$ ) and off-state bias ( $V_{GS} = 0 \text{ V}$ ,  $V_{DS} = 60 \text{ V}$ ). Step irradiations were performed to obtain electrical parameters within 15 min after irradiation at 20, 40, 60, 80 and 100 krad(Si).

TABLE I. MAIN STRUCTURE PARAMETERS

| Description                            | T-MOS                | SGT-MOS              |
|----------------------------------------|----------------------|----------------------|
| Cell pitch( $\mu\text{m}$ )            | 1.5                  | 2.5                  |
| Trench Depth( $\mu\text{m}$ )          | 2                    | 5.5                  |
| Drift layer thickness( $\mu\text{m}$ ) | 8.5                  | 7.5                  |
| N-drift doping( $\text{cm}^{-3}$ )     | $2.5 \times 10^{15}$ | $3.5 \times 10^{16}$ |
| P-body doping( $\text{cm}^{-3}$ )      | $1.9 \times 10^{17}$ | $1.9 \times 10^{17}$ |
| Gate oxide thickness(nm)               | 50                   | 50                   |
| Split-gate oxide thickness(nm)         | -                    | 550                  |
| Isolation oxide thickness(nm)          | -                    | 550                  |

## III. EXPERIMENTAL RESULTS AND DISCUSSION

### A. Threshold Voltage

Fig. 2 shows the measured  $V_{th}$  as a function of dose for SGT-MOS and T-MOS after both on-state and off-state irradiation. The experimental results reveal that a similar decreasing trend in  $V_{th}$  as the dose level increases for both devices. After on-state irradiation,  $V_{th}$  significantly decreases with the increase of TID. However, after off-state irradiation,  $V_{th}$  decreases slightly with an increase in TID. When SGT-MOS is under on-state irradiation, the  $V_{th}$  drops to -4.8 V and the device becomes normally-on at 40 krad(Si). The device remains normally-off at 100 krad(Si) during off-state irradiation. For T-MOS, when under on-state irradiation, the  $V_{th}$  drops to -3.6 V and the device becomes normally-on at 20 krad(Si). Still, this device also remains normally-off at 100 krad(Si) during off-state irradiation.

Fig. 3 shows the schematic band diagram of a MOS structure, illustrating the main processes for radiation-induced charge generation. Electron-hole pairs are created in the oxide by ionizing irradiation, most of the electrons rapidly drift (within picoseconds) toward the gate, while holes move to the Si/SiO<sub>2</sub> interface. As the holes approach the interface, a portion of them will be trapped, forming the positive oxide trapped charges [16]-[18]. Meanwhile, interface trapped charges can also be produced near the Si/SiO<sub>2</sub> interface. Both types of charges are responsible for a negative shift in the threshold voltage.

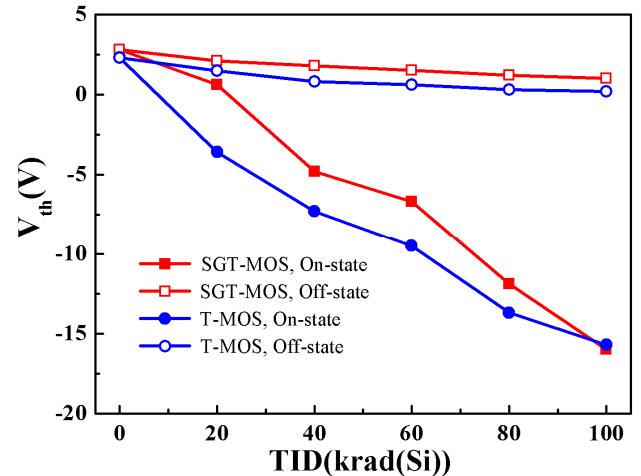


Fig. 2. Measured  $V_{th}$  as a function of dose for SGT-MOS and T-MOS after on-state and off-state irradiation.

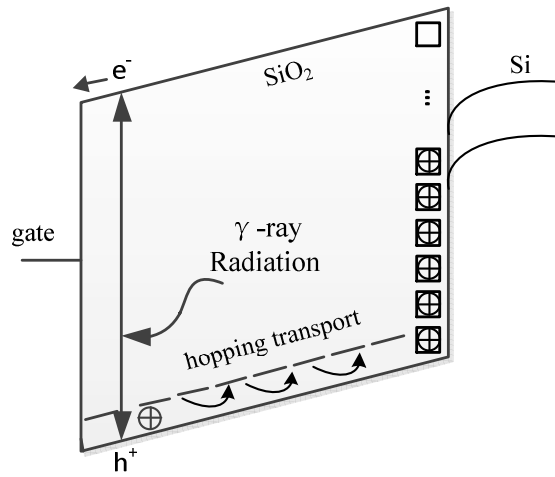


Fig. 3. The schematic band diagram for MOS structure with a positive gate bias, indicating the main processes for radiation-induced charge generation.

The slight discrepancy in  $V_{th}$  shift behavior between SGT-MOS and T-MOS can be attributed to the quality of the oxide layer. The etching and annealing processes during fabrication lead to a poor oxidation environment, which significantly impacts the quality of the gate oxide. Meanwhile, the obvious difference in  $V_{th}$  shift behavior between both types of irradiations can be ascribed to the electric field in the gate oxide. For both SGT-MOS and T-MOS, the  $E_{GOX}$  under on-state irradiation is significantly higher than that under off-state irradiation, leading to more positive trapped charges at the Si/SiO<sub>2</sub> interface and causes a more negative  $V_{th}$ .

### B. Breakdown Voltage

Based on the above analysis, both SGT-MOS and T-MOS become normally-on devices during on-state irradiation when TID reaches 40 krad(Si). To accurately measure the  $BV$  and observe its degradation behavior, the  $V_{GS}$  is set to -20 V for all breakdown voltage tests. For the pre-radiation SGT-MOS, the measured  $BV$  is 108 V, regardless of whether  $V_{GS} = 0$  V or  $V_{GS} = -20$  V. For the pre-radiation T-MOS, the measured  $BV$  is 120 V when  $V_{GS}$  is at 0 V, but drops to 96 V when  $V_{GS}$  is at -20 V, as demonstrated in Fig. 4

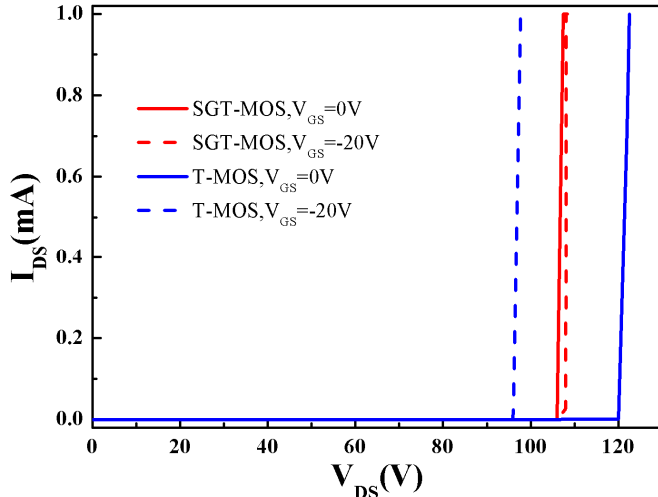


Fig. 4. Experimental breakdown curves at  $V_{GS} = 0$  V or  $V_{GS} = -20$  V for SGT-MOS and T-MOS.

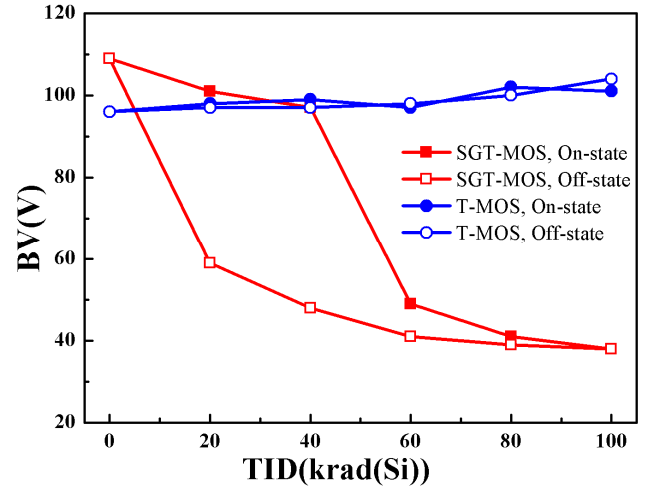


Fig. 5. Measured  $BV$  as a function of dose for SGT-MOS and T-MOS after on-state and off-state irradiation.

Fig. 5 shows the measured  $BV$  as a function of TID dose during on-state and off-state irradiation for both SGT-MOS and T-MOS. It is evident that the  $BV$  displays completely different trends as the dose level increases for both devices. For SGT-MOS, the  $BV$  shows a decreasing trend with an increase in TID after both on-state and off-state irradiation. Hence, it can be inferred that the positive oxide trapped charges in the oxide layer, induced by TID irradiation, are accountable for the  $BV$  shift in SGT-MOS. When the dose accumulates to 100 krad(Si), the  $BV$  drops to approximately 38 V, just 35% of the  $BV$  for fresh devices. Conversely, for T-MOS, the  $BV$  shows a slight increase with an increase in TID, whether after on-state or off-state irradiation. Therefore, TID does not significantly influence the  $BV$  of the T-MOS. When the dose accumulates to 100 krad(Si), the  $BV$  mildly increases to over 101 V for irradiated devices, slightly higher than the 96 V of fresh devices.

In order to analyze the TID irradiation-induced  $BV$  shifting mechanism, irradiation simulation has been performed using Sentaurus TCAD. The positive oxide trapped charges ( $N_{ot}$ ) can be calculated through the following expressions [19], [20], as shown in

$$N_{ot} = g_0 \cdot f_{ot} \cdot f_y \cdot t_{ox} \cdot D \quad (1)$$

$$f_y = \left( \frac{|E_{ox}| + E_0}{|E_{ox}| + E_1} \right)^m \quad (2)$$

where  $g_0$  is  $8.1 \times 10^{12} \text{ cm}^{-3} \text{ rad}(\text{SiO}_2)^{-1}$ ,  $f_y$  is hole yield as a function of the oxide electric field.  $E_0 = 0.1 \text{ V/cm}$ ,  $E_1 = 0.55 \text{ MV/cm}$ , and  $m = 0.7$  for  $\gamma$ -ray Co<sup>60</sup> source.  $E_{ox}$  is the oxide electric field.  $f_{ot}$  is the hole trapping efficiency, the value of which is set as 0.5[14] for SGT-MOS and T-MOS in this work.  $D$  is the dose, and  $t_{ox}$  is the oxide thickness. Significantly, SGT-MOS has a complex trench structure, including gate oxide layer and split-gate oxide layer. Hence, both  $t_{ox}$  and  $E_{ox}$  differ for these two oxide layers. In addition, the interface trapped charges can be neglected under lower TID dose.

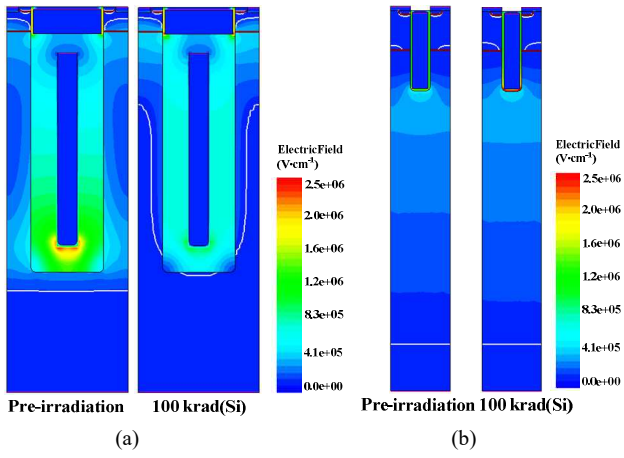


Fig. 6. The electric field distribution profiles at pre-irradiation and 100 krad(Si) for SGT-MOS(a) and T-MOS(b).

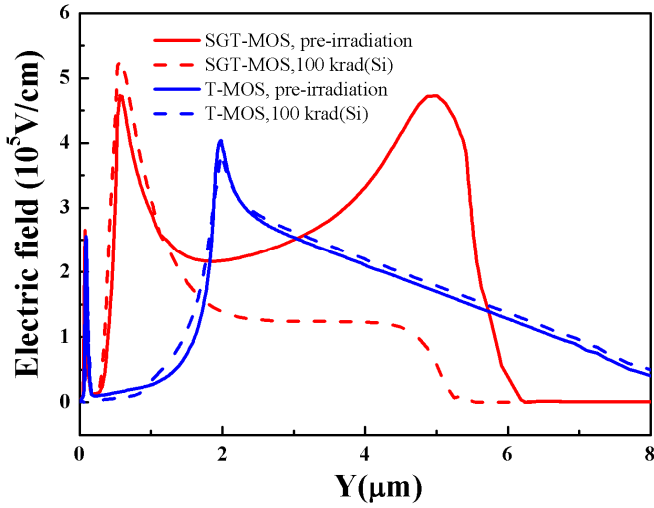


Fig. 7. The electric field distributions along the dielectric trench in the drift region at pre-irradiation and 100 krad(Si) for SGT-MOS and T-MOS.

Fig. 6 illustrates the electric field distribution profiles at pre-irradiation and 100 krad(Si) for SGT-MOS and T-MOS, with the white lines representing the boundaries of the depletion region. For SGT-MOS, when the dose accumulates to 100 krad(Si), the depletion region sharply shrinks, accounting for the significant decrease in  $BV$  after irradiation. However, for T-MOS, as the TID dose increases to 100 krad(Si), the depletion region slightly expands, leading to a minor increase in  $BV$  after irradiation.

To clearly compare the TID irradiation-induced  $BV$  shifting mechanism, the vertical electric field distributions along the dielectric trench in the drift region at pre-irradiation and 100 krad(Si) for both SGT-MOS and T-MOS are obtained, as shown in Fig.7, which are given at approximately  $0.02\mu\text{m}$  from the Si/SiO<sub>2</sub> interface. For SGT-MOS, the electric field develops from a double-humped distribution at pre-irradiation into a single-humped distribution at 100 krad(Si). Meanwhile, the electric field near the N-drift/substrate boundary reduces to a lower value of  $1.2\times 10^5$  V/cm, a decrease of more than 78% relative to the pre-irradiation device. For T-MOS, the peak electric field near the trench bottom decreases to  $3.76\times 10^5$

V/cm from  $4.0\times 10^5$  V/cm of the pre-irradiation device.

Through the above analysis, the differences in  $BV$  shifting behavior between SGT-MOS and T-MOS can be attributed to the electric field redistribution. The positive oxide trapped charges induced by TID could significantly affect the electric field distribution for SGT-MOS, but not for T-MOS.

### C. Improving method

In order to improve the TID tolerance on  $V_{th}$  and  $BV$  for SGT-MOS or T-MOS, three approaches are proposed as follows: a) avoid the high temperature process; b) reduce the oxide layer thickness; c) decrease the drift layer doping.

a) Avoiding the high temperature process is critical for enhancing the radiation hardness of the oxide layer. There are two methods to avoid high temperature process [15], [21]. The first method is to move the P-body annealing ahead of the trench process, which could effectively reduce the oxide defects. The second method is to reduce the oxide growth temperature to below  $1000^\circ\text{C}$ , leading to a higher quality oxide layer. Therefore, these two methods are recommended to improve the TID tolerance owing to fewer oxide traps. Fig. 8 shows the representative radiation-hardened processes for SGT-MOS and T-MOS.

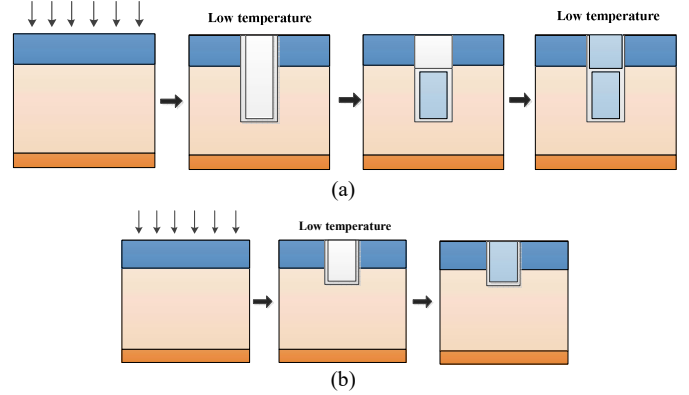


Fig. 8. The representative radiation hardened process for SGT-MOS (a) and T-MOS (b).

b) The TID-induced positive oxide trapped charges have a great influence on the electrical performance for both SGT-MOS and T-MOS. As shown in the  $N_{ot}$  calculation equation, the  $N_{ot}$  is proportional to the oxide layer thickness. Therefore, decreasing the oxide layer thickness could lower the  $N_{ot}$ , which would be an effective way to mitigate both  $V_{th}$  and  $BV$  degradation and further improve the TID tolerance.

c) Decreasing the doping concentration of the drift layer ( $N_d$ ) is a useful method to mitigate the  $BV$  degradation for SGT-MOS due to a wider horizontal depletion region[14]. Fig. 9 illustrates the electric field distributions along the dielectric trench in the drift region at pre-irradiation and 100 krad(Si) with different  $N_d$  for SGT-MOS. For the lower  $N_d$  of  $2.4\times 10^{16}$  cm<sup>-3</sup>, the electric field distribution is more uniform at 100 krad(Si) comparing to that at pre-irradiation. Therefore, a lower  $N_d$  is recommended to improve the TID tolerance for SGT-MOS, despite some sacrifices on the static performances such as  $R_{on,sp}$ .



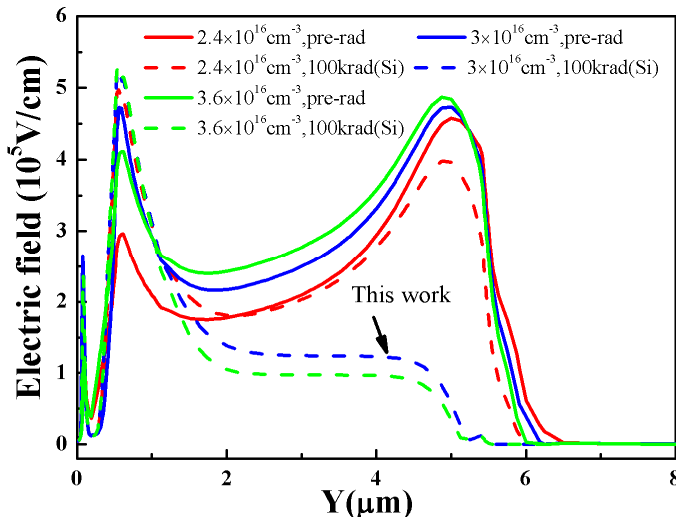


Fig. 9. The electric field distributions along the dielectric trench in the drift region at pre-irradiation and 100 krad(Si) with different  $N_d$  for SGT-MOS.

## I. CONCLUSION

Comparative experimental research was carried out on the  $V_{th}$  and the  $BV$  of SGT-MOS and T-MOS under TID radiation. Experimental results indicate that the  $V_{th}$  decreases with an increase in TID for these two devices. However, the  $BV$  of SGT-MOS displays a decreasing trend as TID increases, while the  $BV$  of T-MOS slightly increases with TID increasing. Therefore, it is necessary to pay attention to the anti-radiation hardening design of SGT-MOS and T-MOS. The findings of this research lay the foundation for the future aerospace application of SGT-MOS and T-MOS.

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